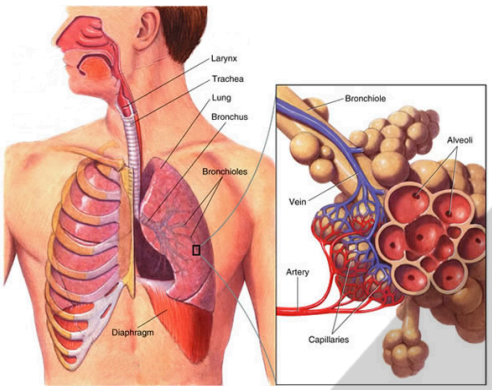
1. Gas Exchange in Humans

1.1. Gas Exchange Surfaces

Properties	Reasons
Thin surface	Short distance to diffuse (one cell thick)
Large surface area	Many molecules can diffuse at once/More alveoli
Good ventilation	Regular fresh air supplies keep up concentration gradients for oxygen and carbon dioxide.
Good blood supply	Gases can be carried to/from the cells that need/produce them
Moist	Allow gases to dissolve, ready for diffusion

1.2. Structure of the Lungs

The lung contains a diaphragm, ribs, intercostal muscles, larynx, trachea, bronchi, bronchioles, alveoli and associated capillaries



- **Cartilage (in the trachea):** prevents the trachea from collapsing during the absence of air and protects it by keeping it open.
- **Ribs:** to protect vital organs and blood vessels and expand and contract (and efficient breathing).
- **Intercostal (internal & external) muscles:** situated between the ribs that create and move the chest wall.
- **Diaphragm:** produces volume and pressure changes in the thorax, leading to the ventilation of the lungs.

Composition of Breathing Dry Air

	Inspired Air	Expired Air
Oxygen	21%	16%
Carbon Dioxide	0.04%	4%
Nitrogen	78%	78%
Water Vapour	Lower	Higher

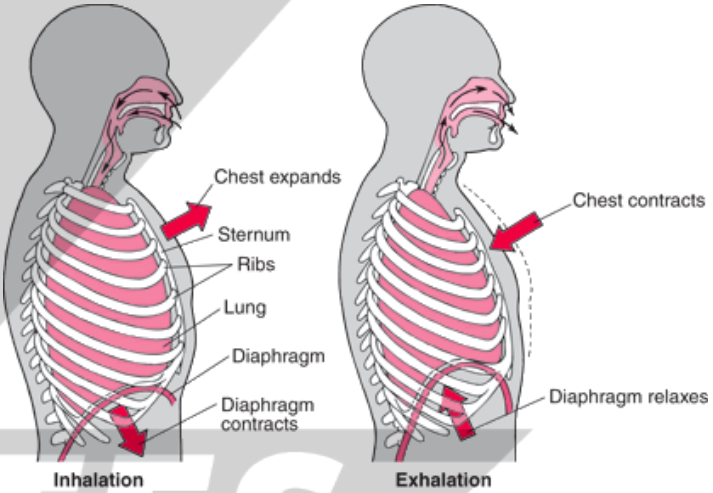
- **Test for CO_2:** Add CO_2 through limewater. +ve result = turns cloudy

1.3. Physical Activity on Breathing

- Physical activity increases the breathing rate – more respiration - and higher CO_2 concentration in the blood.
- This is measured with a spirometer to produce a spirogram.
- During exercise, tissues respire at a higher rate; the change in breathing volume and rate helps keep CO_2 concentration and pH safe.

1.4. Breathing

Inspiration	Expiration
External intercostal muscles contract – pulls ribcage upwards and outwards	External intercostal muscles relax – ribcage falls downwards and inwards
Diaphragm muscles contract – the diaphragm moves downwards, and the volume of the thorax increases	Diaphragm muscles relax – return to a dome shape, and the volume of the thorax decreases
Atmospheric Pressure > Pressure in Thorax	Atmospheric Pressure < Pressure in Thorax
Air moves into the lungs	Air moves out of the lungs



- **Internal intercostal muscles** are used in coughing and sneezing.
- **Mucus & cilia:** goblet cells produce sticky mucus to trap and eliminate particulate matter and microorganisms.
- **Ciliated cells** have cilia, little hairs which sweep/beat back and forward in a coordinated way to brush mucus up the lungs into the mouth.

1.5. Oxygen Debt

- It is the extra oxygen that is needed after anaerobic respiration has taken place, in order to break down the lactic acid produced.

  How it is removed :

- continued fast heart rate to transport lactic acid in blood from the muscles to the liver
- Continued deeper and faster breathing to supply oxygen for aerobic respiration of lactic acid

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- Aerobic respiration of lactic acid in the liver. (the liver breaks down the lactic acid into glucose using oxygen)

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